

Age-related Differences in the Perception of Vowel Length Contrast in Northern Vietnamese: The Case of Hoang Van (Bac Ninh) Variety

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Abstract

Vowel length contrasts in Northern Vietnamese are acoustically distinguished by both duration and spectral cues. This study investigated the perception of long and short vowels in this language and possible differences between older and younger listeners. The results show that, at the group level, listeners in both age groups relied primarily on vowel duration, with F1 and F2 serving as secondary cues. At the individual level, a small subset of older listeners relied mainly on F1. Furthermore, acoustically incongruent stimuli (e.g., long-vowel duration integrated with short-vowel quality) elicited significantly longer response times than congruent stimuli, without significant age-related differences. These findings indicate that both duration and spectral cues contribute to the perception of vowel length in Northern Vietnamese, and the reliance on F1 as the primary cue by some older listeners may indicate a possible ongoing sound change.

Index Terms: sound change, perception, cue weighting, vowel length contrast, Northern Vietnamese

1. Introduction

Phonological contrasts usually rely on multiple acoustic and perceptual cues, which differ in their relative contributions to maintaining the contrast [1]. Previous studies have shown that vowel length contrasts are primarily distinguished by duration, with vowel quality and pitch serving as secondary cues [2, 3, 4, 5, 6, 7, 8]. In many languages, short vowels tend to be more centralized than their long counterparts [9, 10, 11, 12, 13, 14, 15, 16, 17], resulting in differences in vowel quality between long and short vowels.

In Northern Vietnamese (NV), vowel length contrasts occur in closed syllables. Long and short vowels are distinguished by both duration and spectral cues (mainly in vowel height and backness, as reflected by F1 and F2). Because of spectral differences, some researchers have transcribed long and short vowels using different symbols. For example, Emerich [18] used /a/ and /x/ for long vowels, and /e/ and /A/ for their short counterparts. Previous studies on NV have also shown that spectral differences within the vowel length contrast are smaller than duration differences [19, 20]. This suggests that in the production of this multi-cue phonemic contrast, vowel duration carries greater cue weight than vowel quality.

From a sociophonetic perspective, cross-linguistic research has reported a reduction in vowel quality differences among younger speakers in some languages. As a result, these speakers rely more heavily on duration cue [4, 5, 6]. These studies adopt the apparent-time approach [21, 22], which compares speakers from different age groups to identify synchronic variation and

to trace ongoing sound change. As Ohala [23] noted, “sound change is drawn from a pool of synchronic variation.” In other words, sound change emerges from variation in the community, which can be observed in both production and perception. In particular, perceptual variation has been proposed as the initiation of sound change [24, 25, 26]. For this reason, examining age-related differences in perception can provide insight into the mechanisms of sound change.

Some studies have examined the perception of long and short vowels in NV [27, 28]. Nevertheless, these studies did not investigate the relative contribution of different acoustic cues to the perception of vowel length. It also remains unclear whether age-related differences exist in the use of perceptual cues. Furthermore, participants in previous studies were drawn from different speech communities, which may have introduced additional variability. To better understand the perceptual mechanisms underlying vowel length contrasts in NV, and to test for possible differences between younger and older listeners, it is necessary to conduct a more controlled experiment. Such an experiment should examine listeners from different age groups within the same speech community.

The present study investigates the perception of /a(:)/ in NV. We focus on the Hoang Van variety, a rural variety of Vietnamese spoken by approximately 52,900 native speakers in Hoang Van commune, Bac Ninh province, Vietnam. This monolingual speech community is located about 40 km north of the country’s capital. In this study, we address the following questions:

1. How do vowel duration and vowel quality (F1 and F2) contribute to the perception of /a(:)/?
2. Do different integrations of vowel duration and vowel quality influence response time?
3. Do younger and older listeners differ in how they perceive the contrast?

2. Method

2.1. Participants

The experiment was conducted in Hoang Van commune, Bac Ninh province, Vietnam. Data were collected from 41 native speakers of NV living in the commune. The participants were divided into a younger group (n = 20; 11 females; mean age = 24.00 years, SD = 7.20) and an older group (n = 21; 11 females; mean age = 52.52 years, SD = 6.77). All participants and their family members had no long-term experience of living or working outside the commune. None of the participants had received formal training in linguistics or psychology. None reported any speech or hearing disorders.

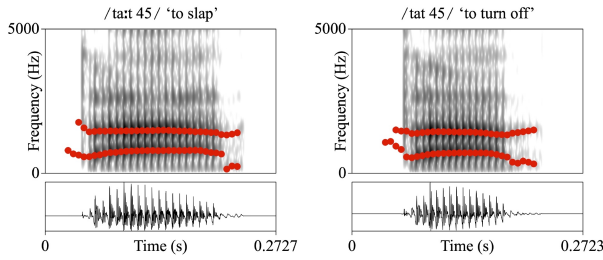


Figure 1: Spectrogram and waveform of the two base sounds, red speckles represent F1 and F2.

2.2. Stimuli

The base sounds /tʌt 45/ ‘to slap’ and /tʌt 45/ ‘to turn off’ were naturally produced by a 42-year-old male speaker. The acoustic properties of the two base sounds are shown in Figure 1 and Table 1. A five-step duration continuum, a three-step F1 continuum, and a three-step F2 continuum were manipulated using Praat v6.4.38 [29]. Formant manipulation was performed using an LPC-based script developed by Winn [30]. The stimuli were normalized for intensity (70 dB), f_0 , and VOT. Finally, 90 target stimuli were obtained (2 base sounds $\times$ 5 duration $\times$ 3 F1 $\times$ 3 F2). In addition, 30 filler items were included, resulting in 120 stimuli in total. The fillers were taken from another project and were included to maintain participants’ attention during the experiment.

Table 1: The acoustic properties of the base sounds

Base	Gloss	Dur (ms)	F1 (Hz)	F2 (Hz)
/tʌt 45/	‘to slap’	158	810	1510
/tʌt 45/	‘to turn off’	120	760	1480

2.3. Procedure

The perception experiment employed a two-alternative forced-choice (2AFC) identification task conducted on Gorilla [31]. Participants completed the experiment in their homes. Stimuli were presented through Sony MDR-7506 headphones. Each stimulus was presented twice in random order, resulting in 240 trials per participant. On each trial, two pictures and their corresponding words were displayed on the screen. Each sound was presented twice, with an inter-repetition interval of 500 ms. After hearing both repetitions, participants were required to respond within 5 seconds by pressing either the Q or P key, they were given breaks after every 30 trials. The experiment took approximately 30–40 minutes to complete for each participant.

Before the main experiment, participants completed a practice session using naturally produced stimuli (the base sounds of the perception experiment). They were allowed to repeat the practice until they were familiar with the procedure.

2.4. Statistical analysis

Trials with response times shorter than 100 ms or longer than 5000 ms were excluded, resulting in the removal of 54 tokens.

A mixed-effects logistic regression model was fitted to the identification data from all speakers using the *glmer()* function in the *lme4* R package [32, 33]. The response variable was the identification response (long vs. short). Fixed effects included

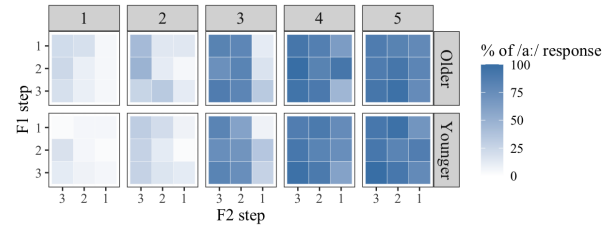


Figure 2: The percentage of ‘long’ response for each stimulus across generations.

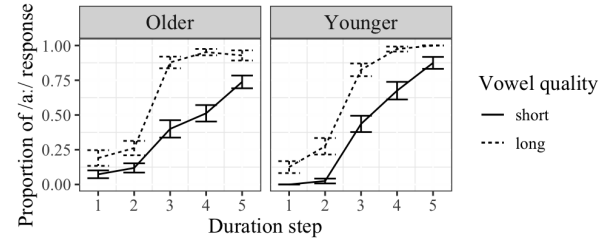


Figure 3: Identification curves of stimuli with duration manipulation only, with the vowel quality of the base sounds maintained.

the continuous step values of vowel duration, F1, F2, and their interactions with generation (younger vs. older, older group was set as the reference level). Random intercepts were included for participants and stimulus repetitions. At the individual level, identification data for each participant were analyzed using logistic regression models fitted with the *glm()* function. The response variable was the identification response, and the predictors were step values of vowel duration, F1, and F2. The relative weight of each cue was indexed by the log-odds coefficient of the corresponding predictor in the model [34].

Response times were log-transformed prior to analysis. A linear mixed-effects regression model was then fitted to the log-transformed response time using the *lmer()* function from the *lmerTest* package [32, 35]. The response variable was log-transformed response time. Fixed effects included stimulus condition (defined as different integrations of vowel duration and vowel quality, as shown in Table 3), generation (younger group was set as the reference level), and their interaction. Random intercepts were included for participants.

3. Result

3.1. Identification data

3.1.1. Group level

The identification results of the two age groups are shown in Figure 2. In this figure, the y-axis and x-axis represent F1 and F2 steps, respectively; larger step values indicate higher F1 and F2 values in the stimuli. The numbers 1–5 indicate duration steps, with larger values corresponding to longer vowel duration. Overall, the two age groups show highly similar perceptual patterns. As vowel duration increases, the proportion of responses identified as the long vowel /a:/ also increases. In addition, vowel quality also has a significant effect: stimuli with vowel quality closer to the long vowel (i.e. higher F1 and F2) are more likely to be perceived as /a:/.

To more clearly illustrate the effects of vowel duration and vowel quality on perception, we further selected stimuli in which vowel quality was held constant and only duration manipulated. The results are shown in Figure 3. In this figure, the y-axis represents the proportion of /a:/ responses, and the x-axis represents duration steps, with larger values indicating longer vowels. Solid lines represent stimuli with short-vowel /a/ quality, dashed lines represent stimuli with long-vowel /a:/ quality, error bars represent standard errors. It can be clearly observed that, when vowel duration is identical, stimuli with long-vowel quality (dashed lines) are consistently more likely to be identified as long vowel than stimuli with short-vowel quality (solid lines). This pattern is consistent across both age groups.

The results of mixed-effects logistic regression model (Table 2) confirm the above observation for the perception of vowel length contrast in NV. In particular, *duration* emerged as the most salient perceptual cue overall ($\beta = 1.22, p < 0.001$), followed by *F1* ($\beta = 0.97, p < 0.001$). *F2* showed the smallest cue weight ($\beta = 0.15, p = 0.004$). All cues had positive log-odds coefficients, indicating that stimuli with longer *duration* and higher *F1* and *F2* were more likely to be identified as /a:/ rather than /a/. The interaction terms between each cue and *generation* capture age-related differences in cue use. As shown in Table 2, no significant age-related differences were found for vowel quality cues. The interactions between *F1* and *generation* ($\beta = -0.03, p > 0.05$) and between *F2* and *generation* ($\beta = 0.10, p > 0.05$) were both non-significant. This indicates that younger and older listeners relied on vowel quality cues in a similar way when identifying long and short vowels. In contrast, a small but significant age-related difference was observed for the *duration* cue. The interaction between *duration* and *generation* was significant ($\beta = 0.25, p < 0.001$), which suggests that younger listeners were more sensitive to *duration* cue than older listeners in vowel length perception.

Table 2: Results of mixed-effects logistic regression model for identification data

	Log-Odds	SE	z value	p
(Intercept)	-5.60	0.25	-22.05	0.000
duration	1.22	0.04	30.80	0.000
F1	0.97	0.06	16.85	0.000
F2	0.15	0.05	2.92	0.004
generation	-0.94	0.37	-2.51	0.012
duration $\times$ generation	0.25	0.06	4.18	0.000
F1 $\times$ generation	-0.03	0.08	-0.33	0.739
F2 $\times$ generation	0.10	0.08	1.27	0.203

Marginal $R^2 = 0.547$ / Conditional $R^2 = 0.578$

To better illustrate age-related differences in the weighting of vowel duration and vowel quality cues, we conducted additional analyses on selected subsets of the perceptual stimuli. We first analyzed stimuli in which duration was manipulated while vowel quality was held constant. *Base* was treated as a binary factor (long vs. short) representing vowel quality. We fitted a mixed-effects logistic regression model with identification response as the dependent variable. Fixed effects included *base* (vowel quality), *generation*, and their interaction. Random intercepts for participants and stimulus repetitions were included. The interaction between *base* and *generation* was visualized in Figure 4 (left), which plots the predicted log-odds of long-vowel responses. Stimuli with long-vowel quality elicited

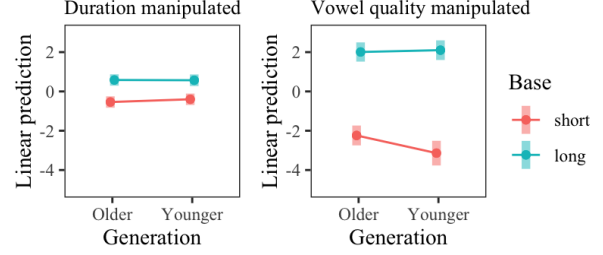


Figure 4: Interaction plots between listener generation and vowel quality (left) and vowel duration (right), with the vertical bars showing 95% confidence interval.

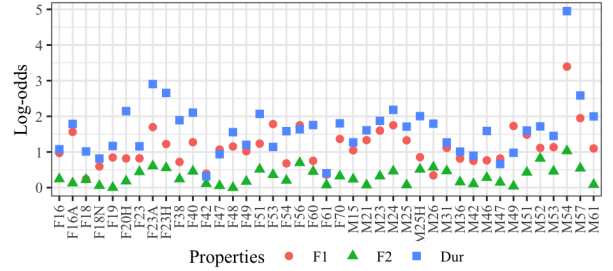


Figure 5: Relative perceptual cue weights by participant. The first letter of each label indicates gender (F = female, M = male); the following two digits indicate age; and the final letter in some participants (e.g., A, N) represents the first letter of their pseudo-first name, used to distinguish individuals of the same age and gender.

more long-vowel responses than those with short-vowel quality in both age groups, with no significant age-related differences observed (*base* $\times$ *generation*: $\beta = -0.15, p > 0.05$).

We then analyzed stimuli in which vowel quality was manipulated while duration was held constant. In this analysis, *base* was treated as a binary factor (long vs. short) representing vowel duration. A mixed-effects logistic regression model was also fitted with the identification response as the response variable. Fixed effects included *base* (vowel duration), *generation* and their interaction. Random intercepts were included for participants and stimulus repetitions. The interaction between *base* and *generation* was visualized in Figure 4 (right), which plots the predicted log-odds of long-vowel responses. Stimuli with long-vowel duration elicited more long-vowel responses than those with short-vowel duration. In addition, younger listeners showed a greater ability to use duration cues to distinguish long and short vowels (*base* $\times$ *generation*: $\beta = 0.99, p = 0.01$).

3.1.2. Individual level

Figure 5 shows the relative perceptual cue weights (higher log-odds values indicate greater weight) for each participant. At the individual level, most listeners relied primarily on duration to distinguish long and short vowels, followed by *F1*, with *F2* showing the smallest cue weight. All younger listeners used duration as the primary perceptual cue. In the older group, however, 6 out of 21 listeners relied primarily on *F1* rather than duration (F42, F47, F53, F56, M47, M49).

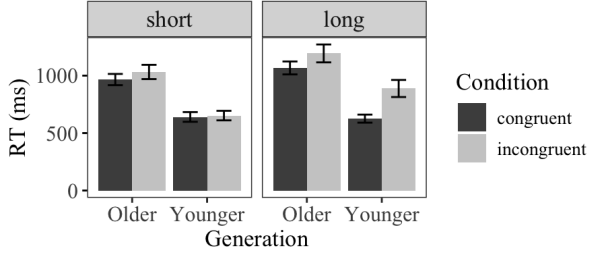


Figure 6: RT in congruent and incongruent conditions by vowel quality (long vs. short) and generation (older vs. younger).

3.2. Response time (RT)

We followed the approach of Gao [36] and Ge et al. [37] to classify the perceptual stimuli and examine whether different integrations of vowel quality and vowel duration affect listeners' response time (RT).

The experiment included five vowel duration steps and three vowel quality steps defined by $F1$ and $F2$. Along the duration continuum, Step 1 was the shortest and Step 5 was the longest. Stimuli at Steps 1–2 were categorized as short, and those at Steps 4–5 as long. Along the $F1$ and $F2$ continua, Step 1 had the lowest $F1$ and $F2$ values, while Step 3 had the highest values. The vowel quality of stimuli with both $F1$ and $F2$ at Step 1 corresponded to /a/, whereas that of stimuli at Step 3 corresponded to /a:/. Only stimuli at duration Steps 1, 2, 4 and 5, with both $F1$ and $F2$ at either Step 1 or Step 3, were included in the analysis.

The selected stimuli were then grouped into four types: (1) short-vowel duration $\times$ short-vowel quality, (2) short-vowel duration $\times$ long-vowel quality, (3) long-vowel duration $\times$ short-vowel quality, and (4) long-vowel duration $\times$ long-vowel quality. Conditions (1) and (4) were classified as *duration-quality congruent*, whereas Conditions (2) and (3) were classified as *duration-quality incongruent*. A summary of this classification is provided in Table 3.

Table 3: Summary of the stimuli conditions in the experiment

Duration step	Vowel quality	Condition
1-2 (short)	/a/	congruent
1-2 (short)	/a:/	incongruent
4-5 (long)	/a/	incongruent
4-5 (long)	/a:/	congruent

Figure 6 shows the mean RT for congruent and incongruent conditions by vowel quality (long vs. short) and generation (older vs. younger), error bars represent standard errors. Overall, the older group exhibited longer RT than the younger group, indicating slower responses in general. This may be due to older individuals tending to process information more slowly. In addition, less familiarity with computer-based tasks may also contribute to longer RT in the older group. Figure 6 further shows that both age groups required longer RT when processing incongruent stimuli than congruent stimuli.

To statistically test these observations, a linear mixed-effects model was fitted with log-transformed RT as the dependent variable. The predictors included stimulus *condition* (congruent vs. incongruent; reference level: congruent),

generation (younger vs. older; reference level: younger), and their interaction. The model results reported in Table 4 support the patterns observed in Figure 6. Specifically, the younger group responded significantly faster than the older group ($\beta = 0.51$, $p < 0.001$), indicating faster perceptual processing. Furthermore, participants showed longer RT for incongruent stimuli than for congruent stimuli ($\beta = 0.15$, $p = 0.001$), without significant age-related differences ($\beta = -0.08$, $p > 0.05$).

Table 4: Results of linear mixed-effects model for response time

	Estimates	SE	t value	p
(Intercept)	6.31	0.08	80.23	0.000
condition	0.15	0.05	3.20	0.001
generation	0.51	0.11	4.61	0.000
condition $\times$ generation	-0.08	0.07	-1.17	0.244
Marginal $R^2 = 0.174$ / Conditional $R^2 = 0.476$				

This response delay suggests that the integration of vowel duration and vowel quality constitutes linguistic knowledge shared by listeners across age groups. When auditory input mismatches this expected integration, additional cognitive processing is required, resulting in slower responses for both younger and older listeners.

4. Discussion and Conclusion

The present study shows that vowel duration, $F1$ and $F2$ all contribute to the perception of vowel length contrast in NV. In particular, vowel duration is the primary perceptual cue, while $F1$ and $F2$ serve as secondary cues. This is consistent with previous findings on the perception of vowel length contrasts in Thai [38], Long'an Zhuang [5] and Daiya [4]. In addition, the response time data indicate that NV listeners have expectations about the integration of vowel duration and vowel quality. When listeners heard incongruent stimuli (for example, long duration integrated with short-vowel quality, or short duration integrated with long-vowel quality), they responded more slowly overall. This effect is similar to findings on the interaction between $f0$ and breathy voice in the perception of Suzhou Wu tones [37]. Taken together, both vowel duration and vowel quality influence vowel length perception in NV, as reflected in both identification rate and processing efficiency.

We also observed age-related differences in the relative weighting of perceptual cues. Specifically, younger listeners showed a higher cue weight for vowel duration than older listeners, indicating greater sensitivity to duration cues. At the individual level, six older participants relied primarily on $F1$ rather than duration to distinguish long and short vowels. These findings suggest that the vowel length contrast in NV is probably undergoing a cue-shifting change, with the primary perceptual cue shifting from vowel quality to duration, making duration cues more salient in the younger group. However, this conclusion awaits further validation in future typological research.

In addition to the perception experiment reported in this study, further analysis of how vowel length contrasts are produced in the same community, along with the inclusion of multiple speakers (e.g., female voices) in the perception experiment, is needed. Such an approach would reveal how vowel length contrasts are produced by speakers and interpreted by listeners across different age groups, thereby shedding light on the perception–production mapping and offering deeper insight into the mechanisms of sound change.

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6. Generative AI Use Disclosure

Two generative AI tools (ChatGPT and Qwen3-Max) were used only to assist with language polishing and correction of grammatical errors. All analyses, interpretations, and conclusions are those of the authors.

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